College Readiness, Staff Demographics, and School Demographics May Predict California High School Graduation Rates*

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California graduation requirements have changed over the years, so one might be interested in what impacts graduation rates. Using data from the Education Data Explorer from the Urban Institute's Center on Education Data and Policy, I investigate whether college readiness, staff demographics, and school demographics can predict graduation rates in California high schools. I first fit a multiple regression model, then a transformed model, then a reduced model using LASSO regularization, which gave the highest prediction accuracy on the validation set. The LASSO model removed Title I eligibility and teacher salary from the model, implying that they are not as influential in predictions. However, there may be more impactful predictors regarding student motivation and parental involvement that can also predict graduation rate.

1 Introduction

The 2025 high school graduation rate for the State of California has risen to 87.5 percent, which is an increase of 4.5 percentage points from 2017 (Thurmond 2025). As graduation requirements have changed and continue to change, it is only natural to question what impacts whether a student will graduate or not. Often these factors are related to the individual student's motivation as well as familial, school, and community factors ("High School Graduation," n.d.). It has already been shown that there are racial and ethnic disparities in high school graduation rates, and parental and teacher involvement, household-income level, reading skill level, and gender affect graduation rates ("High School Graduation," n.d.). These factors can also impact college readiness, such as decisions on taking the SAT or ACT, or stem

*Project repository available at: https://github.com/meganajoseph/261a_project2.

from staff demographics, like teacher salary, and school demographics, such as Title I eligibility. This has motivated the research question: Can college readiness, staff demographics, and school demographics predict graduation rates in California high schools?

As a student pursuing their Masters in Statistics, there are many things I still don't know. I am not very well-versed in education besides having been a student in California and also teaching and tutoring California students which may impact the results of this study. Additionally, I am still learning about regression as a whole which may also impact the study.

In this study, I use the Education Data Explorer from the Urban Institute's Center on Education Data and Policy to compile a dataset including the variables of interest. The Education Data Explorer allows you to create your own dataset based on certain factors, such as school year, location, etc., in order to help students (2025a). This project draws from the data sources Common Core of Data, Small Area Income and Poverty Estimates, The Civil Rights Data Collection, ED*Facts*, Integrated Postsecondary Education Data System, College Scorecard, National Historical Geographic Information System, Federal Student Aid, National Association of College and University Business Officers, National Center for Charitable Statistics, Model Estimates of Poverty in Schools (MEPS), Equity in Athletics Data (EADA), and Campus Safety and Security (2025a).

I used graduation rate as the response and college readiness (the proportion of students in the IB program, the proportion of students in the AP program, the proportion of students in the gifted and talented program, the proportion of students that took one or more AP exams, and the proportion of students taking the SAT or ACT), staff demographics (the number of full-time teachers, the number of full-time first-year teachers, the number of instructional aides and paraprofessionals, the number of counselors, and teacher salary), and school demographics (the number of students enrolled, Title I eligibility, and the number of suspensions) as predictors. The three models created were a multiple regression model, a square root transformed multiple regression model, and a reduced LASSO regularized model.

To pick the best model, I compared the validation root mean squared prediction error of each and picked the one with the lowest which was the LASSO model. This model removed Title I eligibility and teacher salary, implying that these are not as influential in prediction.

The remainder of the paper is structured as follows: Section 2 discusses the data, Section 3 discusses the models, Section 4 presents the results, and Section 5 discusses the implications of the results.

2 Data

The data are from the Education Data Explorer from the Urban Institute's Center on Education Data and Policy. This tool compiles data from national data sources on schools, districts, and colleges in order to assist in creating insights to improve student outcomes (2025a). I used this query to produce a dataset of high schools in California from 2016-2017 with information on school characteristics, absenteeism, college readiness, course offerings, student demographics, discipline, school safety, student outcomes, and teachers and staff (Education Data Portal (Version 0.23.0), Urban Institute 2023). The cleaned dataset has 2125 observations while the original has 6111 observations.

2.1 Common Core of Data

The Common Core of Data (CCD) is the United States Department of Education's database on all public elementary and secondary schools and school districts (2025b). The variables of interest from this dataset are the number of students enrolled and whether or not a school is eligible for participation in either Target Assistance program or school-wide program authorized by Title I of Public Law 103-382.

Student enrollment reports the number of students enrolled in the school. Visualizations show a logarithmic relationship, so a log transformation should be applied, but I will use a square root transformation because of the large number of zeros. A school is Title I eligible if it is in a school district for children from low-income families (2025c). Figure 1 (b) shows that schools that are not Title I eligible have a higher median graduation rate and are less varied, whereas schools that are Title I eligible are much more varied with a larger number of outliers. I picked these variables because small schools have been linked to higher graduation rates, and students from low-income families tend to have higher dropout rates ("High School Graduation," n.d.).

I filtered the data to only include values where the Title I eligibility column was either Yes or No. I also made sure enrollment was greater than zero to account for possible input errors.

2.2 The Civil Rights Data Collection

The Civil Rights Data Collection (CRDC) comes from the United States Department of Education's Office for Civil Rights. This data is collected in order to ensure schools give students equal access to educational opportunities (2025e). Data are collected from public schools and districts, justice facilities, charter schools, alternative schools, and special education schools (2025e).

The key variables of interest from this dataset are the number of students enrolled in the International Baccalaureate Diploma Program, the number of students enrolled in the gifted and talented programs, the number of students enrolled in at least one AP course, the number of students that took one or more AP exams, the number of students participating in the SAT or ACT, the number of suspensions, the number of full-time teachers, number of first-year teachers, number of full-time equivalent instructional aides or paraprofessionals, number of full-time equivalent school counselors, and teacher salary. The number of students enrolled in the International Baccalaureate Diploma Program, the number of students enrolled in the gifted and talented programs, and the number of students enrolled in at least one AP course

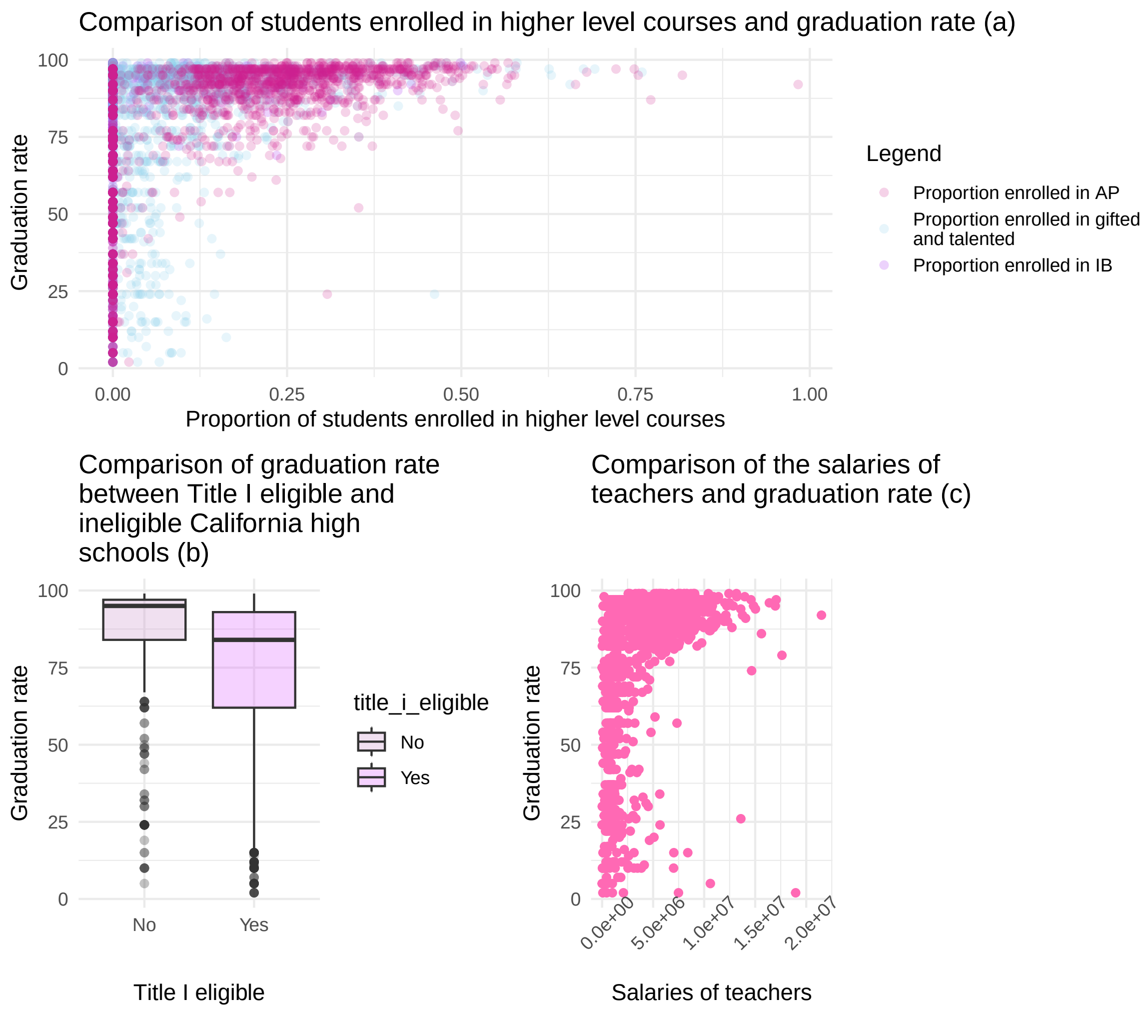

Figure 1: Plot (a) shows a scatter plot showing the relationship between the proportion of students enrolled in high level courses including the IB program (purple), AP program (pink), and gifted and talented programs (blue) (x-axis) and graduation rate (y-axis). Plot (b) shows a boxplot of graduation rate at Title I eligible and ineligible schools. Plot (c) shows the relationship between teacher salary (x-axis) and graduation rate (y-axis).

were transformed to the proportion of students enrolled in each respective program because that is more informative. Figure 1 (a) and (c) show a logarithmic relationship between the proportion of students in higher level courses and graduation rate and teacher salary and graduation rate. The rest of the variables also display this relationship, so a square root transformation is appropriate.

I picked these variables because students participating in higher level courses and/or participating in college-preparatory exams would be most interested in graduating and going to college. Suspensions would make it more difficult for students to graduate. Increased support from experienced teachers and other aides can motivate students to do well in school. Higher teacher salaries would motivate teachers to put more effort in helping their students.

I filtered the data so that the number of suspensions is greater than or equal to zero and that teacher salary is not equal to Not Applicable. Variables regarding the number of students enrolled in any given program were turned into proportions since the number can be impacted by the number of students enrolled.

2.3 ED*Facts*

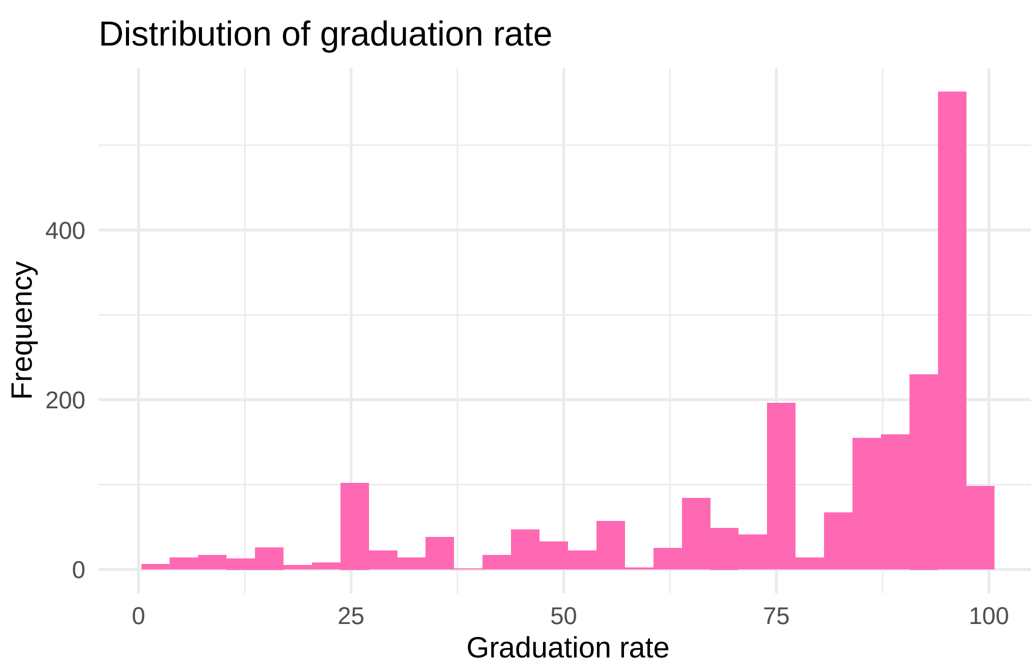

Figure 2: Histogram of graduation rate showing a left skewed distribution.

High school graduation rate comes from the ED*Facts* dataset. Figure 2 shows a left-skewed distribution in graduation rate. It was filtered to remove values that were an empty string or suppressed.

ED*Facts* collects and analyzes data from pre-kindergarten through twelfth grade supplied by state education agencies with other data within the U.S. Department of Education (2025d).

2.4 Alternative Data Sources

A good alternate data source would be at the student level rather than the school level. This can be difficult to find, though, because the students are not legal adults and due to the Family Educational Rights and Privacy Act (FERPA) which protects the distribution of educational records without the consent of parents or the student if they are over 18 years old. Information on absenteeism, family income level, extracurriculars of the students, involvement of parents and teachers, and reading level would also help in predicting graduation rates.

3 Methods

3.1 Base Model

In an attempt to answer this research question, I will first fit a multiple linear regression model using all variables and no transformations. Let Y be the 2125 x 1 vector where each entry represents the graduation rate at each school. Let X be the 2125 x 14 matrix where the first column is the vector of ones and the rest represent one predictor. Let X_1 be the vector representing the number of students enrolled, X_2 be the vector equaling 1 if a school is Title I eligible and 0 if it's not, X_3 be the vector representing the proportion of students enrolled in the International Baccalaureate Diploma Program, X_4 be the vector representing the proportion of students enrolled in the AP program, X_5 be the vector representing the proportion of students enrolled in the gifted and talented program, X_6 be the vector representing the number of students that took one or more AP exams, X_7 be the vector representing the number of students participating in the SAT or ACT, X_8 be the vector representing the number of suspensions, X_9 be the vector representing the number of full-time teachers, X_{10} be the vector representing the number of first-year teachers, X_{11} be the vector representing the number of full-time equivalent instructional aides or paraprofessionals, X_{12} be the vector representing the number of full-time equivalent school counselors, and X_{13} be the vector representing teacher salary.

The model can we written as

$$Y = X\beta + \varepsilon$$

where β is the vector of coefficients for each predictor: β_0 represents the mean of the probability distribution of the graduation rate when all predictors are 0, β_1 represents the average change in graduation rate for an additional student enrolled at at Title I ineligible school, holding all other variables constant, β_2 represents the average change in graduation rate for a Title I school, holding all other predictors constant, β_3 represents the average change in graduation rate for an additional student enrolled in the IB program at at Title I ineligible school, holding all other variables constant, β_4 represents the average change in graduation rate for an additional student enrolled in the AP program at at Title I ineligible school, holding all other variables constant, β_5 represents the average change in graduation rate for an additional student enrolled in the gifted and talented program at at Title I ineligible school, holding all other variables constant, β_6 represents the average change in graduation rate for an additional student that took one or more AP exams at at Title I ineligible school, holding all other variables constant, β_7 represents the average change in graduation rate for an additional student participating in the SAT or ACT at at Title I ineligible school, holding all other variables constant, β_8 represents the average change in graduation rate for an additional suspension instance at at Title I ineligible school, holding all other variables constant, β_9 represents the average change in graduation rate for an additional full-time teacher at at Title I ineligible school, holding all other variables constant, β_{10} represents the average change in graduation rate for an additional first-year teacher at at Title I ineligible school, holding all other variables constant, β_{11} represents the average change in graduation rate for an additional full-time equivalent instructional aide or paraprofessional at at Title I ineligible school, holding all other variables constant, β_{12} represents the average change in graduation rate for an additional full-time equivalent school counselor at at Title I ineligible school, holding all other variables constant, β_{13} represents the average change in graduation rate for a one USD increase in teacher salary at at Title I ineligible school, holding all other variables constant. ε is the vector containing the error.

3.2 Transformed Model

The next model used is the model where each predictor except Title I eligibility is transformed using the square root. While a log transformation would have been ideal, there are many zeros in the dataset which are not part of the domain of a logarithmic function. A square root transformation is similar to a logarithmic one and allows zeros in its domain which makes it a viable substitution.

The model can be written as:

$$Y = Z\beta + \varepsilon$$

where each column $Z_i = \sqrt{X_i}$ if $i \neq 2$ and $Z_i = X_i$ otherwise.

3.3 Least Absolute Shrinkage and Selection Operator (LASSO) Regression Model

The last model created is one using LASSO regression to perform variable selection. LASSO is a shrinkage or regularization regression method that has the ability to "shrink" estimated coefficients to zero, carrying out variable selection. Alongside this, it reduces the variance of parameter estimates. LASSO aims to minimize the equation:

$$\sum_{i=1}^{n}(Y_i - Z_i^T\beta)^2 + \lambda\sum_{j=1}^{p-1}|\beta_j|$$

where $\lambda \geq 0$ is a tuning parameter that controls the level of regularization, n represents the number of observations, and p is the number of predictors. Essentially, we are minimizing the error sum of squares (SSE) with a penalty term added to it.

When picking the best lambda value, `cv.glmnet()` uses cross-validation to pick the best lambda that produces a mean squared error within one standard error of the mean squared error of the true best lambda (Friedman, Hastie, and Tibshirani 2010a). This is because the true best lambda may overfit the model by setting many estimated coefficients to zero. The user is able to pick any lambda used by the function based on their own analysis. For ease, I will use the default best lambda.

3.4 Model Selection

To determine which model is the best, I will first split the data into a 70% training, 15% testing, and 15% validation set. Then, I will train the models on the training set and find the root mean squared prediction error using the validation set. This is defined as

$$\sqrt{E[(Y^* - \hat{Y}^*)^2]}$$

where Y^* is the true graduation rate and $\hat{Y}^*$ is the predicted graduation rate using the model. This is the square root of the expected squared difference between the prediction and the true value over the validation data population. The one that gives the best root mean squared error will be the best model. Finally, I will use the testing set to estimate the performance for the final model.

3.5 Assumptions of Linear Regression

The assumptions of using linear regression for inference are:

1. The variables used accurately reflect the quantities of interest, the model should include all predictors, and the model should be applicable for the research question.
2. The sample data is representative of the population of interest.
3. The mean function must be correct.
4. Errors must be independent.
5. The variance of the errors must be equal.
6. Errors must be normal.

Since the goal of this analysis is point prediction, the most important assumption that should be met is linearity. If my goal was to create prediction intervals, I would need to meet the normality of errors assumption. After the transformations, there appears to be a linear relationship between the predictors and graduation rate.

3.6 Limitations of Analysis

One limitation of this analysis is that it is at the school level rather than the student level. While there are many reasons as to why this is the case, it does limit the analysis, and predictions can only be made for schools. An analysis at the student level, though, would change the problem from linear regression to logistic regression since the response would be whether the student graduated or not. Another limitation is that interaction effects are not included in any models. This is because it reduces the interpretability even though it could improve performance. In a future analysis, researching which variables have a theoretical interaction with each other would build upon this. Additionally, there may be models better suited for this research question that I have not learned yet.

3.7 Software

I use the `lm()` function from the R programming language (R Core Team 2025) to create the base model and the transformed model. I use the `cv.glmnet()` function from the `glmnet` package (Friedman, Hastie, and Tibshirani 2010b) to use LASSO regression to select the most important variables.

4 Results

The final model is the LASSO regression model. Table 1 summarizes the parameter estimates. Each slope estimate can be interpreted as the average change in graduation rate for a unit increase in the given variable, holding all other variables constant. The proportion of students in the AP program, taking one or more AP exams, and participating in the SAT or ACT have the highest estimates which means they have a larger effect and could be due to differences in scaling. Since it is not possible for all predictors to be zero, the intercept term has little meaning. Title I eligibility and teacher salary were removed in the process, resulting in an estimated parameter of 0.

Table 1: Table showing the estimated parameters of the LASSO model

Slope Estimate	Column	Value
$\hat{\beta}_0$	Intercept	57
$\hat{\beta}_1$	Number of students enrolled	0.867
$\hat{\beta}_2$	Tile I eligible	0
$\hat{\beta}_3$	Number of suspensions	-0.087
$\hat{\beta}_4$	Proportion of students in the IB program	9.447

Slope Estimate	Column	Value
$\hat{\beta}_5$	Proportion of students in the gifted and talented program	-2.677
$\hat{\beta}_6$	Proportion of students in the AP program	51.519
$\hat{\beta}_7$	Proportion of students taking one or more AP exams	-17.214
$\hat{\beta}_8$	Proportion of students participating in the SAT or ACT	22.371
$\hat{\beta}_9$	Number of full-time teachers	-3.15
$\hat{\beta}_{10}$	Number of full-time first-year teachers	1.552
$\hat{\beta}_{11}$	Number of instructional aides or paraprofessionals	-0.907
$\hat{\beta}_{12}$	Number of counselors	-0.707
$\hat{\beta}_{13}$	Teacher salary	0

Since LASSO regularization removed Title I eligibility and teacher salary from the model, it believed it was not as important in predicting graduation rate or may already be explained by some other variables. It's surprising that many estimated parameters are negative when one would assume that, for example, having more students taking an AP exam would increase graduation rates rather than decrease. There may be additional factors at play behind the scenes, especially since this is at the school level.

4.1 Estimating Performance

Table 2 presents the root mean squared prediction error for each model created and the final model, with the base model having the highest RMSPE and the LASSO model having the lowest RMSPE, and the MSPE on the test set is comparable to that of the validation set for the same model.

Table 2: This table shows the root mean squared prediction errors on the validation set for each created model and for the chosen model on the testing set.

Base Model RMSPE (Validation)	Transformed Model RMSPE (Validation)	LASSO Model RMSPE (Validation)	Final Model RMSPE (Test)
19.448	18.654	18.377	18.397

The performance for each model, however, is comparable, and none perform very well since the average error is about 18 percentage points for the best model. This is significant because it is a large gap, but it's most likely due to graduation being more nuanced and explained by variables not tracked by schools

5 Discussion

The aim of this paper was to answer the following research question: Can college readiness, staff demographics, and school demographics predict graduation rates in California high schools? I investigated this using data from the Common Core of Data, The Civil Rights Data Collection, and ED*Facts*. Then, I created three models. The first was a multiple regression model with graduation rate as the response and the number of students enrolled, Title I eligibility, the proportion of students in the IB program, the proportion of students in the AP program, the proportion of students in the gifted and talented program, the proportion of students that took one or more AP exams, the proportion of students taking the SAT or ACT, the number of suspensions, the number of full-time teachers, the number of full-time first-year teachers, the number of instructional aides and paraprofessionals, the number of counselors, and teacher salary as predictors. The second model transformed each predictor except Title I eligibility using a square root transformation. The last model was made using LASSO on the previous model to select the best predictors. The model that predicted the best out of the three was the LASSO model:

$$Y = \begin{bmatrix} | & | & | & | & | & | & | & | & | & | & | & | & | & | \\ 1 & Z_1 & Z_2 & Z_3 & Z_4 & Z_5 & Z_6 & Z_7 & Z_8 & Z_9 & Z_{10} & Z_{11} & Z_{12} & Z_{13} \\ | & | & | & | & | & | & | & | & | & | & | & | & | & | \end{bmatrix} \begin{bmatrix} 57 \\ 0.867 \\ 0 \\ -0.087 \\ 9.447 \\ -2.677 \\ 51.519 \\ -17.214 \\ 22.371 \\ -3.15 \\ 1.552 \\ -0.907 \\ -0.707 \\ 0 \end{bmatrix}$$

The predictors with a parameter estimate of zero, Title I eligibility and teacher salary, were removed by LASSO regularization. This model has a test root mean squared prediction error of 18.397. This means that college readiness, staff demographics, and school demographics can predict graduation rates, but there may be other factors that can help increase prediction accuracy.

5.1 Potential Weaknesses

One weakness of this study is the lack of interaction terms. For example, knowing if a school is Title I eligible or not may impact the proportion of students who take an AP exam or SAT/ACT because one needs to pay to take those tests. Including them could explain more variability and increase the predictive power of the final model.

Another weakness is that there are more factors to graduation rate than the predictors I chose. Many of these factors, like family income level, if the student is first generation, and student responsibilities, are at the student level and not generally available or quantifiable. Graduation rates can also be affected by sudden changes, such as a death in the family, that may cause a decrease in performance or even cause the student to drop-out altogether.

5.2 Possible Extensions

An extension of this study would be to put out a survey at the student level to find more information on the students' motivations and other factors mentioned above. Another extension would be to answer an inference-based research question, such as using an F-test to determine if any of the chosen predictors do not have a relationship with graduation rate. One could also predict success in college and beyond using a different set of predictors, which is useful for college admissions.

References

2025a. *Education Data Explorer*. https://educationdata.urban.org/data-explorer/about.

———. 2025b. *Common Core of Data (CCD) - Common Core of Data (CCD)*. National Center for Education Statistics. https://nces.ed.gov/ccd/.

———. 2025c. *National Center for Education Statistics (NCES)*. U.S. Department of Education. https://nces.ed.gov/fastfacts/display.asp?id=158.

———. 2025d. *U.S. Department of Education*. https://www.ed.gov/data/edfacts-initiative.

———. 2025e. *Ocrdata.ed.gov*. https://ocrdata.ed.gov/.

Education Data Portal (Version 0.23.0), Urban Institute. 2023. "Common Core of Data; the Civil Rights Data Collection; Model Estimates of Poverty in Schools (MEPS); EDFacts." https://educationdata.urban.org/documentation/.

Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. 2010b. "Regularization Paths for Generalized Linear Models via Coordinate Descent." *Journal of Statistical Software* 33 (1): 1–22. https://doi.org/10.18637/jss.v033.i01.

———. 2010a. "Regularization Paths for Generalized Linear Models via Coordinate Descent." *Journal of Statistical Software* 33 (1): 1–22. http://www.jstatsoft.org/.

"High School Graduation." n.d. *Office of Disease Prevention and Health Promotion*. https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health/literature-summaries/high-school-graduation.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Thurmond, Tony. 2025. "California Students Show Continued Increase in Academic Achievement and Graduation Rates, Reduction in Chronic Absenteeism with Release of 2025 School Dashboard." *California Department of Education News Release*. https://www.cde.ca.gov/nr/ne/yr25/yr25rel49.asp.